

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
APPLIED BUSINESS ANALYTICS

Practical – II

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1. Introduction to Business Analytics:

a.Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest.

Code:-

```
import pandas as pd
df = pd.read_csv("supermarket_sales - supermarket_sales.csv")
print(df.head())
```

Output:-

```
Invoice ID Branch City Customer type Gender \
0 750-67-8428 A Yangon Member Female
1 226-31-3081 C Naypyitaw Normal Female
2 631-41-3108 A Yangon Normal Male
3 123-19-1176 A Yangon Member Male
4 373-73-7910 A Yangon Normal Male

Product line Unit price Quantity Tax 5% Total Date \
0 Health and beauty 74.69 7 26.1415 548.9715 1/5/2019
1 Electronic accessories 15.28 5 3.8200 80.2200 3/8/2019
2 Home and lifestyle 46.33 7 16.2155 340.5255 3/3/2019
3 Health and beauty 58.22 8 23.2880 489.0480 1/27/2019
4 Sports and travel 86.31 7 30.2085 634.3785 2/8/2019

Time Payment cogs gross margin percentage gross income Rating
0 13:08 Ewallet 522.83 4.761905 26.1415 9.1
1 10:29 Cash 76.40 4.761905 3.8200 9.6
2 13:23 Credit card 324.31 4.761905 16.2155 7.4
3 20:33 Ewallet 465.76 4.761905 23.2880 8.4
4 10:37 Ewallet 604.17 4.761905 30.2085 5.3
```

Code:-

```
sales = df["Tax 5%"]
print("Mean :", sales.mean())
print("Median :", sales.median())
print("Mode :")
print(sales.mode())
print("Minimum :", sales.min())
print("Maximum :", sales.max())
print("Range :", sales.max() - sales.min())
print("Variance :", sales.var())
print("Standard Deviation :", sales.std())
print("Count :", sales.count())
print("Sum :", sales.sum())
print("\nDescriptive Statistics")
print(sales.describe())
```

Output:-

```
Output exceeds the size limit. Open the full output data in a text editor
Mean : 15.379368999999999
Median : 12.088000000000001
Mode :
0 4.1540
1 4.4640
2 8.3770
3 9.0045
4 10.3260
5 10.3635
6 12.5700
7 13.1880
8 22.4280
9 39.4800

Name: Tax 5%, dtype: float64
Minimum : 0.5085
Maximum : 49.65
Range : 49.1415
Variance : 137.09659414448348
Standard Deviation : 11.708825480998659
Count : 1000
Sum : 15379.368999999999

Descriptive Statistics
count 1000.000000
mean 15.379369
...
50% 12.088000
75% 22.445250
max 49.650000
Name: Tax 5%, dtype: float64
```

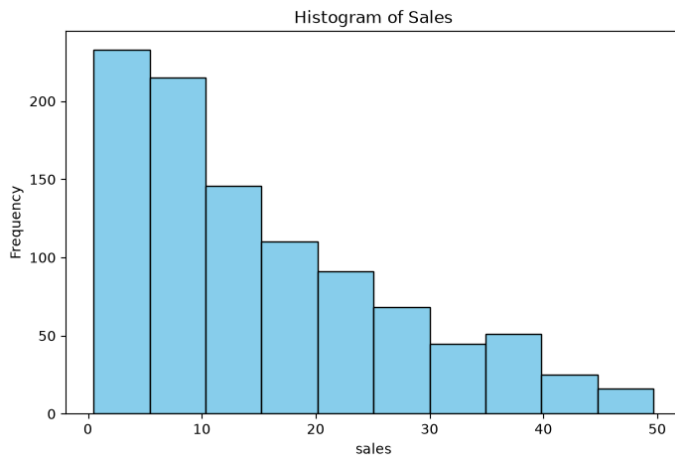
b.Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

Code:-

```
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(8,5))
plt.hist(sales, bins=10, color="skyblue", edgecolor="black")
plt.title("Histogram of Sales")
plt.xlabel("sales")
plt.ylabel("Frequency")
plt.show()
```

[58] ✓ 0.2s

Output:-



Code:-

```
plt.figure(figsize=(8,3))
sns.boxplot(x=sales, color="orange")
plt.title("Box Plot of Sales")
plt.xlabel("Area_sqft")
plt.show()
```

[59] ✓ 0.1s

Output:-

